JEEVAN JAYASURIYA

Madison, WI, USA
Email: jayasuriyaar@wisc.edu| Phone: (315) 261-3235 | Website: jeevanrj.github.io

SUMMARY

Ph.D. student in Industrial and Systems Engineering specializing in neuroergonomics, biomechanics, multimodal machine learning, and human behavior modeling. Background in mind–motor–machine interaction and embodied cognition, with experience in multimodal physiological sensing, temporal modeling, and latent behavioral state analysis using active inference concepts and TCNs. Interested in affective computing and human-AI interaction.

EDUCATION

• **University of Wisconsin–Madison**	Ph.D., Industrial & Systems Engineering	Expected August 2027
• **Clarkson University**	M.S., Mechanical Engineering	2023
• **The Open University of Sri Lanka**	B.Tech. (Hons.), Mechatronics Engineering	2017
• **University of Sri Jayewardenepura**	B.S. (Hons.), Physics	2015

EXPERIENCE

UNIVERSITY OF WISCONSIN–MADISON — Madison, Wisconsin

Graduate Research Assistant – NeuroErgonomics Lab 2024–Present

- Collaborate with NASA Johnson Space Center to investigate how sensory perturbations, workload, and fatigue influence human behavior, adaptation, and decision-making in safety-critical human–machine systems.
- Design and execute long-duration human-subject experiments under sensory and cognitive stressors, integrating behavioral, physiological, and temporal interaction data.
- Analyze multimodal datasets including fNIRS, EMG, eye tracking, kinematics, heart rate, and force plate signals to model human adaptation, latent behavioral states, and performance dynamics in complex operational environments.

CLARKSON UNIVERSITY — Potsdam, New York

Graduate Research Assistant – Astronautics and Robotics Lab 2022–2023

- Applied musculoskeletal simulation and human movement analysis to support adaptive exoskeleton design and evaluation, focusing on human adaptation, movement strategies, and human–system interaction.
- Integrated human-centered design, biomechanical analysis, and mechatronic development to analyze posture, movement, and task performance in physically demanding operational environments.

PRIOR PROFESSIONAL EXPERIENCE

- University Lecturer – Embedded Systems (see website for teaching modules)
- Engineering Trainee – Robotics, Automation, PLC Systems (Sri Lanka)

SELECTED PUBLICATIONS

- *Ergonomics* (2026): Firefighter motion capture & musculoskeletal simulation–driven biomechanical analysis (article)
- *Int. J. of Coal Science & Technology* (2022)*:* Machine vision-based automated fire-control system (article)

TECHNICAL SKILLS

• **Programming & Statistics:**	Python (PyTorch, TensorFlow, scikit-learn), C++, MATLAB, R
• **Multimodal human sensing:**	fNIRS, EMG, ECG/HR, Eye Tracking, Motion Capture, Force Plates
• **Simulation & Embedded Tools:**	AnyBody, OpenSim, Simulink, SolidWorks, PTC Creo, Sensors & actuators, MCUs
• **Ergonomics & Human Factors:**	3DSSPP, NASA-TLX, POMS, KSS, RULA, REBA, NIOSH

AWARDS & AFFILIATIONS

- Best Student Work Award in the Aerospace Systems Technical Group (ASTG) in ASPIRE (2024 & 2025)
- Academic Excellence Awards in Engineering (2016 & 2017)
- Student Affiliate Human Factors and Ergonomics Society (HFES) (2024–present)
- Associate Member, Institute of Engineers Sri Lanka (IESL) (2018–present)